Capítulo 1

Decimal → Número-Máquina

-52.234375

1. $(52)_{10} = (110100)_2$

$52 \div 2 = 26$, resto 0; $26 \div 2 = 13$, resto 0; $13 \div 2 = 6$, resto 1; $6 \div 2 = 3$, resto 0; $3 \div 2 = 1$, resto 1; $1 \div 2 = 0$, resto 1

2. $(0.234375)_{10} = (0.001111)_2$

$0.234375 \times 2 = 0.46875$
$0.46875 \times 2 = 0.9375$
$0.9375 \times 2 = 1.875$
$0.875 \times 2 = 1.75$
$0.75 \times 2 = 1.5$
$0.5 \times 2 = 1.0$
$0 \times 2 = 0$

3. $(52.234375)_{10} = (110100.001111)_2 =$
$= (1.10100001111)_2 \times 2^5$

4. Precisão Simples:

$c - 127 = 5 \Leftrightarrow c = 132$

$(132)_{10} = (10000100)_2$

$132 \div 2 = 66$, resto 0; $66 \div 2 = 33$, resto 0; $33 \div 2 = 16$, resto 1; $16 \div 2 = 8$, resto 0; $8 \div 2 = 4$, resto 0; $4 \div 2 = 2$, resto 0; $2 \div 2 = 1$, resto 0; $1 \div 2 = 0$, resto 1

-52 negativo
positivo = 0

$[1 \quad \underbrace{10000100}_{8 \text{ dígitos}} \quad \underbrace{10100001111 0 \ldots 0}_{23 \text{ dígitos}}]_2$

Precisão Dupla:

$c - 1023 = 5 \Leftrightarrow c = 1028$

$(1028)_{10} = (10000000100)_2$

$1028 \div 2 = 514$, resto 0; $514 \div 2 = 257$, resto 0; $257 \div 2 = 128$, resto 1; $128 \div 2 = 64$, resto 0; $64 \div 2 = 32$, resto 0; $32 \div 2 = 16$, resto 0; $16 \div 2 = 8$, resto 0; $8 \div 2 = 4$, resto 0; $4 \div 2 = 2$, resto 0; $2 \div 2 = 1$, resto 0; $1 \div 2 = 0$, resto 1

$[1 \quad \underbrace{10000000100}_{11 \text{ dígitos}} \quad \underbrace{10100001111 0 \ldots 0}_{52 \text{ dígitos}}]_2$

$[(-1)^s \quad \text{Expoente} \quad \text{Mantissa}]$

Número-Máquina → Decimal

$[\,1 \quad 10000100 \quad 10100000111110\ldots0\,]_2$

① 1 → Negativo

0 → Positivo

② $(10000100)_2 = 1\times2^7 + 1\times2^2 = (\underbrace{132}_{c})_{10}$

$c - 127 = \gamma \Leftrightarrow 132 - 127 = \gamma \Leftrightarrow \gamma = \underline{5}$

③ $(1.1010000001111)_2 \times 2^5 = (110100\,.\,001111)$

④ $(110100)_2 = 1\times2^5 + 1\times2^4 + 1\times2^2 = 32+16+4 = (\underline{52})_{10}$

⑤ $(0.001111)_2 = 1\times2^{-3} + 1\times2^{-4} + 1\times2^{-5} + 1\times2^{-6} = 0.125 + 0.0625 + 0.03125 + 0.015625 =$

$= (\underline{0.234375})_{10}$

⑥ -52.234375

Erros

$\varepsilon_a = |x - \tilde{x}|$

$\varepsilon_r = \dfrac{|x - \tilde{x}|}{|x|}$

Algarismos Significativos

$x = 0.02136 \qquad \tilde{x} = 0.02147 \qquad \beta = 10$

Maior inteiro não-negativo m para o qual se tem:

$\varepsilon_r \leq 5\times10^{-m}$

$$\frac{|0.02136 - 0.02147|}{|0.02136|} \simeq 0.0051498 \leq 0.5\times10^{1-2} = 5\times10^{-2}$$

2 algarismos significativos

Capítulo 3

Método de Gauss Simples

$$\begin{cases} 3x_1 + 4x_2 + 3x_3 = 10 \\ x_1 + 5x_2 - x_3 = 7 \\ 6x_1 + 3x_2 + 7x_3 = 15 \end{cases}$$

$$\left[\begin{array}{ccc|c} \underline{3} & 4 & 3 & 10 \\ \underline{1} & 5 & -1 & 7 \\ \underline{6} & 3 & 7 & 15 \end{array}\right] \longrightarrow \begin{array}{l} m_{21} = \frac{1}{3} \quad 1 - m \times 3 = 0 \Leftrightarrow m = \frac{1}{3} \\ m_{31} = \frac{6}{3} = 2 \end{array}$$

$$\begin{array}{r} \longrightarrow \\ L_2 - \frac{1}{3}L_1 \\ L_3 - 2L_1 \end{array} \left[\begin{array}{ccc|c} 3 & 4 & 3 & 10 \\ 0 & \underline{11/3} & -2 & 11/3 \\ 0 & \underline{-5} & 1 & -5 \end{array}\right] \longrightarrow \begin{array}{l} m_{32} = -5 - m \times \frac{11}{3} = 0 \Leftrightarrow \\ \Leftrightarrow m = -\frac{15}{11} \end{array}$$

$$\begin{array}{r} \longrightarrow \\ L_3 + \frac{15}{11}L_2 \\ \end{array} \left[\begin{array}{ccc|c} 3 & 4 & 3 & 10 \\ 0 & 11/3 & -2 & 11/3 \\ 0 & 0 & -19/11 & 0 \end{array}\right]$$

$$\begin{cases} 3x_1 + 4x_2 + 3x_3 = 10 \\ \frac{11}{3}x_2 - 2x_3 = \frac{11}{3} \\ -\frac{19}{11}x_3 = 0 \end{cases} \Leftrightarrow \begin{cases} x_1 = 2 \\ x_2 = 1 \\ x_3 = 0 \end{cases}$$

Método de Gauss com pivot parcial

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 11 \\ -2 & 8 & -1 & -15 \\ 4 & -6 & 5 & 29 \end{array}\right] \xrightarrow[L_1 \leftrightarrow L_3]{} \left[\begin{array}{ccc|c} 4 & -6 & 5 & 29 \\ -2 & 8 & -1 & -15 \\ 1 & -3 & 2 & 11 \end{array}\right] \longrightarrow$$

Maior |a| da coluna 1

$m_{21} = -2 - m \times 4 = 0 \Leftrightarrow$
$\Leftrightarrow m = -\frac{1}{2}$

$m_{31} = 1 - m \times 4 = 0 \Leftrightarrow$
$\Leftrightarrow m = \frac{1}{4}$

$$\xrightarrow[\substack{L_2 + \frac{1}{2}L_1 \\ L_3 - \frac{1}{4}L_1}]{} \left[\begin{array}{ccc|c} 4 & -6 & 5 & 29 \\ 0 & 5 & 3/2 & -\frac{1}{2} \\ 0 & -\frac{3}{2} & \frac{3}{4} & \frac{15}{4} \end{array}\right] \longrightarrow$$

Maior |a| da coluna 2

$m_{32} = -\frac{3}{2} - m \times 5 = 0 \Leftrightarrow m = -\frac{3}{10}$

$$\xrightarrow[L_3 + \frac{3}{10}L_2]{} \left[\begin{array}{ccc|c} 4 & -6 & 5 & 29 \\ 0 & 5 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & \frac{6}{5} & \frac{18}{5} \end{array}\right]$$

$$\begin{cases} 4x_1 - 6x_2 + 5x_3 = 29 \\ 5x_2 + \frac{3}{2}x_3 = -\frac{1}{2} \\ \frac{6}{5}x_3 = \frac{18}{5} \end{cases} \Leftrightarrow \begin{cases} x_1 = 2 \\ x_2 = -1 \\ x_3 = 3 \end{cases}$$

Fatorização de Doolittle

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}}_{L} \times \underbrace{\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}}_{U} = \underbrace{\begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & -1 & 1 \end{bmatrix}}_{A}$$

$u_{11} = 3$

$l_{21} \times u_{11} = 0 \Leftrightarrow l_{21} \times 3 = 0 \Leftrightarrow l_{21} = 0$

$l_{31} \times u_{11} = 1 \Leftrightarrow l_{31} \times 3 = 1 \Leftrightarrow l_{31} = \frac{1}{3}$

$u_{12} = 1$

$l_{21} \times u_{12} + u_{22} = 2 \Leftrightarrow 0 \times 1 + u_{22} = 2 \Leftrightarrow u_{22} = 2$

$l_{31} \times u_{12} + l_{32} \times u_{22} = -1 \Leftrightarrow \frac{1}{3} \times 1 + l_{32} \times 2 = -1 \Leftrightarrow l_{32} = -\frac{4}{3} \div 2 \Leftrightarrow l_{32} = -\frac{2}{3}$

$u_{13} = 0$

$l_{21} \times u_{13} + u_{23} = 1 \Leftrightarrow u_{23} = 1$

$l_{31} \times u_{13} + l_{32} \times u_{23} + u_{33} = 1 \Leftrightarrow \frac{1}{3} \times 0 + \left(-\frac{2}{3}\right) \times 1 + u_{33} = 1 \Leftrightarrow u_{33} = \frac{5}{3}$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{3} & -\frac{2}{3} & 1 \end{bmatrix} \qquad U = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & \frac{5}{3} \end{bmatrix}$$

Fatorização de Crout

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \times \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$\underbrace{}_{L} \qquad \underbrace{}_{U} \qquad \underbrace{}_{A}$$

$l_{11} = 3$

$l_{21} \times 1 = 0 \Leftrightarrow l_{21} = 0$

$l_{31} \times 1 = 1 \Leftrightarrow l_{31} = 1$

—

$l_{11} \times u_{12} = 1 \Leftrightarrow 3 \times u_{12} = 1 \Leftrightarrow u_{12} = \frac{1}{3}$

$l_{21} \times u_{12} + l_{22} = 2 \Leftrightarrow 0 \times \frac{1}{3} + l_{22} = 2 \Leftrightarrow l_{22} = 2$

$l_{31} \times u_{12} + l_{32} = -1 \Leftrightarrow 1 \times \frac{1}{3} + l_{32} = -1 \Leftrightarrow l_{32} = -\frac{4}{3}$

—

$l_{11} \times u_{13} = 0 \Leftrightarrow 3 \times u_{13} = 0 \Leftrightarrow u_{13} = 0$

$l_{21} \times u_{13} + l_{22} \times u_{23} = 1 \Leftrightarrow 0 \times 0 + 2 \times u_{23} = 1 \Leftrightarrow u_{23} = \frac{1}{2}$

$l_{31} \times u_{13} + l_{32} \times u_{23} + l_{33} = 1 \Leftrightarrow 1 \times 0 + \left(-\frac{4}{3}\right) \times \frac{1}{2} + l_{33} = 1 \Leftrightarrow l_{33} = \frac{5}{3}$

$$L = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 1 & -\frac{4}{3} & \frac{5}{3} \end{bmatrix} \qquad U = \begin{bmatrix} 1 & \frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}$$

Matriz Inversa com LU

$L\vec{y} = \vec{b} \rightarrow U\vec{x} = \vec{y}$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{3} & -\frac{2}{3} & 1 \end{bmatrix} \quad U = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 5/3 \end{bmatrix}$$

$\vec{b} = (1, 0, 0)$

$L\vec{y} = \vec{b}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{3} & -\frac{2}{3} & 1 \end{bmatrix} \times \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{3} \end{bmatrix}$$

$U\vec{x} = \vec{y}$

$$\begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & \frac{5}{3} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{3} \end{bmatrix} \Leftrightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \boxed{\begin{bmatrix} \frac{3}{10} \\ \frac{1}{10} \\ -\frac{1}{5} \end{bmatrix}}$$ 1ª coluna A^{-1}

Repetir

$\vec{b} = (0, 1, 0)$

... 2ª coluna A^{-1}

$\vec{b} = (0, 0, 1)$

... 3ª coluna A^{-1}

$$\underline{\underline{A^{-1}}} = \begin{bmatrix} \frac{3}{10} & -\frac{1}{10} & \frac{1}{10} \\ \frac{1}{10} & \frac{3}{10} & -\frac{3}{10} \\ -\frac{1}{5} & \frac{2}{5} & \frac{3}{5} \end{bmatrix}$$

Decomposição de Cholesky

$$\begin{bmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{bmatrix} \times \begin{bmatrix} L_{11} & L_{21} & L_{31} \\ 0 & L_{22} & L_{32} \\ 0 & 0 & L_{33} \end{bmatrix} = \begin{bmatrix} 4 & -3 & 0 \\ -3 & 6 & 2 \\ 0 & 2 & 3 \end{bmatrix}$$

$(L_{11})^2 = 4 \Leftrightarrow \underline{L_{11} = 2}$

$L_{21} \times L_{11} = -3 \Leftrightarrow 2 \times L_{21} = -3 \Leftrightarrow \underline{L_{21} = -\frac{3}{2}} \Leftrightarrow L_{21} \simeq -1.5$

$L_{31} \times L_{11} = 0 \Leftrightarrow \underline{L_{31} = 0}$

$L_{11} \times L_{21} = -3 \Leftrightarrow 2 \times \left(-\frac{3}{2}\right) = -3 \checkmark$

$(L_{21})^2 + (L_{22})^2 = 6 \Leftrightarrow \frac{9}{4} + (L_{22})^2 = 6 \Leftrightarrow \underline{L_{22} = \frac{\sqrt{15}}{2}} \Leftrightarrow L_{22} \simeq 1.936$

$L_{31} \times L_{21} + L_{32} \times L_{22} = 2 \Leftrightarrow 0 \times \left(-\frac{3}{2}\right) + \frac{\sqrt{15}}{2} \times L_{32} = 2 \Leftrightarrow \underline{L_{32} = \frac{4\sqrt{15}}{15}}$

$L_{32} \simeq 1.033$

...

$(L_{31})^2 + (L_{32})^2 + (L_{33})^2 = 3 \Leftrightarrow 0^2 + \left(\frac{4\sqrt{15}}{15}\right)^2 + (L_{33})^2 = 3 \Leftrightarrow \underline{L_{33} = \sqrt{\frac{435}{225}}}$

$L_{33} \simeq 1.390$

$$\underline{\underline{L}} = \begin{bmatrix} 2 & 0 & 0 \\ -\frac{3}{2} & \frac{\sqrt{15}}{2} & 0 \\ 0 & \frac{4\sqrt{15}}{15} & \sqrt{\frac{435}{225}} \end{bmatrix}$$

Método de Gauss-Seidel

$$\begin{cases} 3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \\ 0.1x_1 + 7x_2 - 0.3x_3 = -19.3 \\ 0.3x_1 - 0.2x_2 + 10x_3 = 71.4 \end{cases}$$

$$T_{GS} = -(D+L)^{-1} \times U$$

$$x_1^{(K)} = \frac{1}{3}\left(7.85 + 0.1x_2^{(K-1)} + 0.2x_3^{(K-1)}\right)$$

$$x_2^{(K)} = \frac{1}{7}\left(-19.3 - 0.1x_1^{(K-1)} + 0.3x_3^{(K-1)}\right)$$

$$x_3^{(K)} = \frac{1}{10}\left(71.4 - 0.3x_1^{(K-1)} + 0.2x_2^{(K-1)}\right)$$

Se $x_1^{(0)} = x_2^{(0)} = x_3^{(0)} = 0$

1ª iteração:

$x_1^{(1)} = 2.61667$

$x_2^{(1)} = -2.75714$

$x_3^{(1)} = 7.14$

2ª iteração:

$x_1^{(2)} = 3.00076$

$x_2^{(2)} = -2.48852$

$x_3^{(2)} = 7.00636$

$$\varepsilon_{r1}^{(2)} = \frac{|x_1^{(1)} - x_1^{(2)}|}{|x_1^{(1)}|} = 0.05610$$

$$\varepsilon_{r2}^{(2)} = 0.09743$$

$$\varepsilon_{r3}^{(2)} = 0.01872$$

Método de Gauss-Jacobi:

$$\begin{cases} \boxed{3}x_1 - x_2 = 8 \\ x_1 + \boxed{6}x_2 = 9 \end{cases} \qquad e \leq 0.01$$

$$D = \begin{bmatrix} 3 & 0 \\ 0 & 6 \end{bmatrix} \quad D^{-1} = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{6} \end{bmatrix} \quad L = \begin{bmatrix} 3 & 0 \\ 1 & 6 \end{bmatrix} \quad U = \begin{bmatrix} 3 & -1 \\ 0 & 6 \end{bmatrix}$$

Matriz iteradora

$\underline{T_J} = -D^{-1}(L+U) =$

$$= \begin{bmatrix} -\frac{1}{3} & 0 \\ 0 & -\frac{1}{6} \end{bmatrix} \times \begin{bmatrix} 6 & -1 \\ 1 & 12 \end{bmatrix} = \begin{bmatrix} -2 & \frac{1}{3} \\ -\frac{1}{6} & -2 \end{bmatrix}$$

—— // ——

$$D^{-1}\vec{b} = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{6} \end{bmatrix} \begin{bmatrix} 8 \\ 9 \end{bmatrix} = \begin{bmatrix} \frac{8}{3} \\ 3/2 \end{bmatrix}$$

$$x^{(k)} = \begin{bmatrix} -2 & \frac{1}{3} \\ -\frac{1}{6} & -2 \end{bmatrix} x^{(k-1)} + \begin{bmatrix} \frac{8}{3} \\ \frac{3}{2} \end{bmatrix}$$

—— // ——

$$x_1^{(k)} = \frac{1}{3}\left(8 + x_2^{(k-1)}\right) \qquad |3| > |-1| \checkmark$$

$$x_2^{(k)} = \frac{1}{6}\left(9 - x_1^{(k-1)}\right) \qquad |6| > |1| \checkmark$$

Matriz estritamente de diagonal dominante

$x_1^{(0)} = x_2^{(0)} = 0$

$x_1^{(1)} = \frac{8}{3}$, $x_2^{(1)} = \frac{3}{2}$ $\qquad \|x^{(1)} - x^{(0)}\| = \sqrt{(8/3)^2 + (3/2)^2} = \frac{\sqrt{337}}{6} \simeq 3.05959 > e$

$x_1^{(2)} = \frac{19}{6}$, $x_2^{(2)} = \frac{19}{18}$ $\qquad \|x^{(2)} - x^{(1)}\| = \sqrt{(19/6 - 8/3)^2 + (19/18 - 3/2)^2} = \sqrt{(1/2)^2 + (4/9)^2} = \frac{\sqrt{145}}{18} \simeq 0.66898 > e$

$x_1^{(3)} = \frac{163}{54}$, $x_2^{(3)} = \frac{35}{36}$ $\qquad \|x^{(3)} - x^{(2)}\| = \sqrt{(4/27)^2 + (1/12)^2} \simeq 0.16998 > e$

$x_1^{(4)} = \frac{323}{108}$, $x_2^{(4)} = \frac{323}{324}$ $\qquad \|x^{(4)} - x^{(3)}\| = \sqrt{(1/36)^2 + (2/81)^2} \simeq 0.03717 > e$

$x_1^{(5)} = \frac{2915}{972}$, $x_2^{(5)} = \frac{649}{648}$ $\qquad \|x^{(5)} - x^{(4)}\| = \sqrt{(2/243)^2 + (1/216)^2} \simeq 0.00944 < e$

Logo, $x_1 \simeq 3$ e $x_2 \simeq 1$

$$A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Método de Newton

$$x^{(K+1)} = x^{(K)} - [J_F(x^{(K)})]^{-1} \times F(x^{(K)})$$

$$J_F(x^{(K)}) \times H^{(K)} = -F(x^{(K)})$$

$$x^{(K-1)} = x^{(K)} + H^{(K)}$$

$$\begin{cases} x_2 - x_1^2 = 0 \\ x_1 x_2 - x_1 - 1 = 0 \end{cases} \qquad x^{(0)} = (1.3 \,,\; 1.8) \qquad e < 10^{-2} = 0.01$$

Derivar

$$F(x) = \begin{bmatrix} x_2 - x_1^2 \\ x_1 x_2 - x_1 - 1 \end{bmatrix} \qquad J_F(x) = \begin{bmatrix} -2x_1 & 1 \\ x_2 - 1 & x_1 \end{bmatrix}$$

$$F(x^{(0)}) = \begin{bmatrix} 0.11 \\ 0.04 \end{bmatrix} \qquad J_F(x^{(0)}) = \begin{bmatrix} -2.6 & 1 \\ 0.8 & 1.3 \end{bmatrix}$$

$$[J_F(x^{(0)})]^{-1} = -\frac{1}{4.18}\begin{bmatrix} 1.3 & -1 \\ -0.8 & -2.6 \end{bmatrix} =$$

$$= \begin{bmatrix} -0.31100 & 0.23923 \\ 0.19139 & 0.62201 \end{bmatrix}$$

$$\begin{bmatrix} x_1^{(1)} \\ x_2^{(1)} \end{bmatrix} = \begin{bmatrix} 1.3 \\ 1.8 \end{bmatrix} - \begin{bmatrix} -0.31100 & 0.23923 \\ 0.19139 & 0.62201 \end{bmatrix} \times \begin{bmatrix} 0.11 \\ 0.04 \end{bmatrix} =$$

$$= \begin{bmatrix} 1.3 \\ 1.8 \end{bmatrix} - \begin{bmatrix} -0.02464 \\ 0.04593 \end{bmatrix} = \begin{bmatrix} 1.32464 \\ 1.75407 \end{bmatrix} \qquad \|x^{(1)} - x^{(0)}\| \simeq 0.05212 > e$$

$$F(x^{(1)}) = \begin{bmatrix} -0.00060 \\ -0.00113 \end{bmatrix} \qquad J_F(x^{(1)}) = \begin{bmatrix} -2.64928 & 1 \\ 0.75407 & 1.32464 \end{bmatrix}$$

$$[J_F(x^{(1)})]^{-1} = -\frac{1}{4.26}\begin{bmatrix} 1.32464 & -1 \\ -0.75407 & -2.64928 \end{bmatrix} =$$

$$= \begin{bmatrix} -0.31095 & 0.23474 \\ 0.17701 & 0.62180 \end{bmatrix}$$

$$\begin{bmatrix} x_1^{(2)} \\ x_2^{(2)} \end{bmatrix} = \begin{bmatrix} 1.32464 \\ 1.75407 \end{bmatrix} - \begin{bmatrix} -0.31095 & 0.23474 \\ 0.17701 & 0.62190 \end{bmatrix} \times \begin{bmatrix} -0.00060 \\ -0.00113 \end{bmatrix} =$$

$$= \begin{bmatrix} 1.32464 \\ 1.75407 \end{bmatrix} - \begin{bmatrix} -0.00008 \\ -0.00081 \end{bmatrix} = \begin{bmatrix} 1.32472 \\ 1.75488 \end{bmatrix} \qquad \|x^{(2)} - x^{(1)}\| \simeq 0.00081 < e$$

Logo, $x_1 \simeq 1.32472$ e $x_2 \simeq 1.75488$

Capítulo 4

Forma de Lagrange do polinómio interpolador

	x_0	x_1	x_2
x	0.5	1.2	3.2
y	-3.2	1.8	-1.9
	$f(x_0)$	$f(x_1)$	$f(x_2)$

3 pontos → Grau 2

$$P_2(x) = L_{2,0}(x)\times f(x_0) + L_{2,1}(x)\times f(x_1) + L_{2,2}(x)\times f(x_2)$$

$$P_3(x) = L_{3,0}(x)\times f(x_0) + L_{3,1}(x)\times f(x_1) + L_{3,2}(x)\times f(x_2) + L_{3,3}(x)\times f(x_3)$$

$$L_{2,0}(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-1.2)(x-3.2)}{((0.5)-1.2)(0.5-3.2)} = \frac{1}{1.89}(x-1.2)(x-3.2)$$

$$L_{2,1}(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-0.5)(x-3.2)}{(1.2-0.5)(1.2-3.2)} = -\frac{1}{1.4}(x-0.5)(x-3.2)$$

$$L_{2,2}(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-0.5)(x-1.2)}{(3.2-0.5)(3.2-1.2)} = \frac{1}{5.4}(x-0.5)(x-1.2)$$

$$P_2(x) = \frac{1}{1.89}(x-1.2)(x-3.2)\times(-3.2) + \left(-\frac{1}{1.4}\right)(x-0.5)(x-3.2)\times 1.8 +$$

$$+\frac{1}{5.4}(x-0.5)(x-1.2)(-1.9) =$$

$$= -\frac{3.2}{1.89}(x^2-3.2x-1.2x+3.84) - \frac{1.8}{1.4}(x^2-3.2x-0.5x+1.6) +$$

$$+\left(-\frac{1.9}{5.4}\right)(x^2-1.2x-0.5x+0.6) =$$

$$= -1.69312x^2 + 7.44974x - 6.50159 - 1.28571x^2 + 4.75714x - 2.05714 -$$

$$-0.35185x^2 + 0.59815x - 0.21111 =$$

$$P_2(x) = -3.33068x^2 + 12.80503x - 8.76984$$

$$P_2(1.5) = 2.943675$$

Polinómio de Neville

$p_{i,i}(x) = y_i$

$$p_{i,j}(x) = \frac{x - x_i}{x_j - x_i} \times p_{i+1,j} - \frac{x - x_j}{x_j - x_i} \times p_{i,j-1}$$

	0	1	2	3	4
x	0.1	0.5	1.0	0.2	0.6
$f(x)$	0.100167	0.521095	1.175201	0.201336	0.636654

Aproximação para $f(0.3)$:

$$\left.\begin{aligned} p_{0,0}^{*} &= 0.100167 \\ p_{1,1}^{*} &= 0.521095 \\ p_{2,2}^{*} &= 1.175201 \\ p_{3,3}^{*} &= 0.201336 \\ p_{4,4}^{*} &= 0.636654 \end{aligned}\right\} \; {}^{*}\, p_{\cdot,\cdot}(0.3)$$

$$\begin{array}{lllll} p_{0,0} & & & & \\ p_{1,1} & p_{0,1} & & & \\ p_{2,2} & p_{1,2} & p_{0,2} & & \\ p_{3,3} & p_{2,3} & p_{1,3} & p_{0,3} & \\ p_{4,4} & p_{3,4} & p_{2,4} & p_{1,4} & p_{0,4} \end{array}$$

$$p_{0,1}(0.3) = \frac{0.3-0.1}{0.5-0.1} \times 0.521095 - \frac{0.3-0.5}{0.5-0.1} \times 0.100167 = 0.310631$$

$$p_{1,2}(0.3) = \frac{0.3-0.5}{1.0-0.5} \times 1.175201 - \frac{0.3-1.0}{1.0-0.5} \times 0.521095 = 0.259453$$

$$p_{2,3}(0.3) = \frac{0.3-1}{0.2-1} \times 0.201336 - \frac{0.3-0.2}{0.2-1} \times 1.175201 = 0.323069$$

$$p_{3,4}(0.3) = \frac{0.3-0.2}{0.6-0.2} \times 0.636654 - \frac{0.3-0.6}{0.6-0.2} \times 0.201336 = 0.310166$$

$$p_{0,2}(0.3) = \frac{0.3-0.1}{1-0.1} \times 0.259453 - \frac{0.3-1}{1-0.1} \times 0.310631 = 0.299258$$

$p_{1,3}(0.3) = 0.301864$

$p_{2,4}(0.3) = 0.300489$

$p_{0,3}(0.3) = 0.304470$

$p_{1,4}(0.3) = 0.304614$

$p_{0,4}(0.3) = \underline{0.304528}$

Forma de Newton do polinómio interpolador com diferenças divididas

	x_k	$f[x_k]$	$f[\cdot,\cdot]$	$f[\cdot,\cdot,\cdot]$	$f[\cdot,\cdot,\cdot,\cdot]$	$f[\cdot,\cdot,\cdot,\cdot,\cdot]$
0	0.1	0.100167	1.052320	0.284324	0.186130	0.020800
1	0.5	0.521095	1.308212	0.302937	0.196530	
2	1.0	1.175201	1.217331	0.322590		
3	0.2	0.201336	1.088295			
4	0.6	0.636654				

$$f[x_0,x_1]=\frac{f[x_0]-f[x_1]}{x_0-x_1}=\frac{0.100167-0.521095}{0.1-0.5}=1.052320$$

$f[x_1,x_2]=1.308212$

$f[x_2,x_3]=1.217331$

$f[x_3,x_4]=1.088295$

$$f[x_0,x_1,x_2]=\frac{f[x_0,x_1]-f[x_1,x_2]}{x_0-x_2}=\frac{1.052320-1.308212}{0.1-1}=0.284324$$

$f[x_1,x_2,x_3]=0.302937$

$f[x_2,x_3,x_4]=0.322590$

$$f[x_0,x_1,x_2,x_3]=\frac{f[x_0,x_1,x_2]-f[x_1,x_2,x_3]}{x_0-x_3}=0.186130$$

$f[x_1,x_2,x_3,x_4]=0.196530$

$f[x_0,x_1,x_2,x_3,x_4]=0.020800$

$$P_4(x)=0.100167+1.052320(x-x_0)+0.284324(x-x_0)(x-x_1)+$$
$$+0.186130(x-x_0)(x-x_1)(x-x_2)+$$
$$+0.020800(x-x_0)(x-x_1)(x-x_2)(x-x_3)$$

Aproximação para $f(0.3)$ e $f(0.9)$:

$P_4(0.3)=0.304528$

$P_4(0.9)=1.026585$

Diferenças progressivas e regressivas de Newton

k	x_k	$f(x_k)$	Δf	$\Delta^2 f$	$\Delta^3 f$	$\Delta^4 f$
0	0	0				
			0.201358			
1	0.2	0.201358		0.008801		
			0.210159		0.013024	
2	0.4	0.411517		0.021825		0.016961
			0.231984		0.029985	
3	0.6	0.643501		0.051810		
			0.283794			
4	0.8	0.927295				
		0	1	2	3	4

• Progressivas

• Regressivas

$\Delta \rightarrow$ diferenças progressivas $\qquad \nabla \rightarrow$ diferenças regressivas

$\Delta f_0 = f_1 - f_0 = 0.201358 - 0 = 0.201358$

$\Delta f_1 = f_2 - f_1 = 0.210159$

$\Delta f_2 = f_3 - f_2 = 0.231984$

$\Delta f_3 = f_4 - f_3 = 0.283794$

$\Delta^2 f_0 = \Delta f_1 - \Delta f_0 = 0.008801$

$\Delta^2 f_1 = \Delta f_2 - \Delta f_1 = 0.021825$

$\Delta^2 f_2 = \Delta f_3 - \Delta f_2 = 0.051810$

$\Delta^3 f_0 = \Delta^2 f_1 - \Delta^2 f_0 = 0.013024$

$\Delta^3 f_1 = \Delta^2 f_2 - \Delta^2 f_1 = 0.029985$

$\Delta^4 f = \Delta^3 f_1 - \Delta^3 f_0 = 0.016961$

Diferenças progressivas (Aproximação $f(0.13)$):

$s = \dfrac{0.13 - 0}{0.2} = 0.65$ $+$ $\qquad h = \dfrac{0.8 - 0}{4} = 0.2 \qquad h = \dfrac{x_n - x_0}{n}$

$P_4(x) = 0 + 0.201358 \cdot \dfrac{s}{1!} + 0.008801 \cdot \dfrac{s(s-1)}{2!} + 0.013024 \cdot \dfrac{s(s-1)(s-2)}{3!} + \qquad s = \dfrac{x - x_0}{h}$

$+ 0.016961 \cdot \dfrac{s(s-1)(s-2)(s-3)}{4!}$

$P_4(0.13) = 0.117437$

Diferenças Regressivas (Aproximação $f(0.67)$):

$s = \dfrac{0.67 - 0.8}{0.2} = -0.65$ $-$

$P_4(x) = 0.927295 + 0.283794 \cdot \dfrac{s}{1!} + 0.051810 \cdot \dfrac{s(s+1)}{2!} + 0.029985 \cdot \dfrac{s(s+1)(s+2)}{3!} +$

$+ 0.016961 \cdot \dfrac{s(s+1)(s+2)(s+3)}{4!}$

$P_4(0.67) = 0.734891$

Método dos Mínimos Quadrados (Regressão Linear e Quadrática)

Regressão Linear

$$\tilde{y} = a_0 + a_1 x$$

$$a_1 = \frac{n \times \sum x_i y_i - \sum x_i \times \sum y_i}{n \times \sum x_i^2 - (\sum x_i)^2}$$

$$a_0 = \overline{y_i} - \overline{x_i} \times a_1$$

x	1	2	4	5	7	8	10
y	1.0	2.0	4.0	4.0	5.0	6.0	7.0

$$\sum x_i = 1+2+4+5+7+8+10 = 37$$

$$\sum y_i = 1.0+2.0+4.0+4.0+5.0+6.0+7.0 = 29$$

$$\sum x_i y_i = 1+2\times2+4\times4+5\times4+7\times5+8\times6+10\times7 = 194$$

$$\sum x_i^2 = 1+2^2+4^2+5^2+7^2+8^2+10^2 = 259$$

$$(\sum x_i)^2 = 37^2 = 1369$$

$$\overline{x_i} = \frac{37}{7} \qquad \overline{y_i} = \frac{29}{7}$$

Alternativa

$$\begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

$$a_1 = \frac{7 \times 194 - 37 \times 29}{7 \times 259 - 1369} = \frac{95}{148} \qquad a_0 = \frac{29}{7} - \frac{37}{7} \times \frac{95}{148} = \frac{3}{4}$$

$$\tilde{y} = 0.75 + 0.64189x$$

$$E(a_0, a_1) = \sum [y_i - (a_0 + a_1 \times x_i)]^2$$

$$E(a_0, a_1) = \sum [y_i - (0.75 + 0.64189 x_i)]^2 = 0.722973$$

Regressão Quadrática

$$\tilde{y} = a_0 + a_1 . x + a_2 . x^2$$

$$n.a_0 + (\sum x_i).a_1 + (\sum x_i^2).a_2 = \sum y_i$$

$$(\sum x_i).a_0 + (\sum x_i^2).a_1 + (\sum x_i^3).a_2 = \sum x_i y_i$$

$$(\sum x_i^2).a_0 + (\sum x_i^3).a_1 + (\sum x_i^4).a_2 = \sum x_i^2 y_i$$

$$\sum x_i^3 = 2053$$

$$\sum x_i^4 = 17395$$

$$\sum x_i^2 y_i = 1502$$

$$\begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \sum x_i^2 y_i \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 7 & 37 & 259 & 29 \\ 37 & 259 & 2053 & 194 \\ 259 & 2053 & 17395 & 1502 \end{array}\right] \xrightarrow{\substack{L_2 - \frac{37}{7}L_1 \\ L_3 - 37L_1}} \left[\begin{array}{ccc|c} 7 & 37 & 259 & 29 \\ 0 & \frac{444}{7} & 684 & \frac{285}{7} \\ 0 & 684 & 7812 & 429 \end{array}\right] \xrightarrow{L_3 - \frac{399}{37}L_2} \left[\begin{array}{ccc|c} 7 & 37 & 259 & 29 \\ 0 & \frac{444}{7} & 684 & \frac{285}{7} \\ 0 & 0 & \frac{16128}{37} & -\frac{372}{37} \end{array}\right]$$

$$a_2 = -0.023065$$

$$a_1 = 0.890620$$

$$a_0 = 0.288699$$

$$E(a_0, a_1, a_2) = \sum [y_i - (a_0 + a_1 . x_i + a_2 . x_i^2)]^2$$

$$\tilde{y} = 0.288699 + 0.890620x - 0.023065x^2$$

Reta que Minimiza Erro

$$f(y) = a_0 + a_1 y$$

$$\begin{bmatrix} n & \sum y_i \\ \sum y_i & \sum y_i^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} \sum x_i \\ \sum x_i y_i \end{bmatrix}$$

Capítulo 5

Diferenças finitas progressivas e centradas

Diferença finita progressiva

$$f'(x_0) \simeq \frac{f(x_0+h)-f(x_0)}{h}$$

$f(x) = \sqrt{1+x^2}$ $\quad f'(0.5)$ $\quad h = 0.1, 0.05, 0.025$

$$f'(0.5) \simeq \frac{f(0.5+0.1)-f(0.5)}{0.1} \simeq 0.481564$$

$$f'(0.5) \simeq \frac{f(0.5+0.05)-f(0.5)}{0.05} \simeq 0.464745$$

$$f'(0.5) \simeq \frac{f(0.5+0.025)-f(0.5)}{0.025} \simeq 0.456068$$

Derivação

$$f'(x_0) = \lim_{h\to 0} \frac{f(x_0+h)-f(x_0)}{h}$$

$$f'(x_0) \simeq \frac{f(x_0+h)-f(x_0)}{h}$$

Diferença finita regressiva

$$f'(x_0) \simeq \frac{f(x_0)-f(x_0-h)}{h}$$

Diferença finita centrada

$$f'(x) \simeq \frac{f(x+h)-f(x-h)}{2h}$$

$$f'(0.5) \simeq \frac{f(0.5+0.1)-f(0.5-0.1)}{2\times 0.1} \simeq 0.445787$$

$$f'(0.5) \simeq \frac{f(0.5+0.05)-f(0.5-0.05)}{2\times 0.05} \simeq 0.446856$$

$$f'(0.5) \simeq \frac{f(0.5+0.025)-f(0.5-0.025)}{2\times 0.025} \simeq 0.447124$$

Fórmulas de derivação dos 3 pontos

$f(0.1) = 0.041393$ $\quad f(0.2) = 0.079181$ $\quad f(0.3) = 0.113943$

Diferença progressiva (3 pontos)

$$f'(x) \simeq \frac{-3\times f(x)+4f(x+h)-f(x+2h)}{2h}$$

$$f'(0.1) \simeq \frac{-3\times 0.041393+4\times 0.079181-0.113943}{2\times 0.1} \simeq 0.39301$$

Diferença regressiva (3 pontos)

$$f'(x) \simeq \frac{3f(x)-4f(x-h)+f(x-2h)}{2h}$$

$$f'(0.3) \simeq \frac{3\times f(0.3)-4f(0.2)+f(0.1)}{2\times 0.1} \simeq 0.33249$$

Diferença finita centrada = Diferença finita centrada (3 pontos)

$$f'(0.2) \simeq \frac{f(0.3)-f(0.1)}{2\times 0.1} = 0.36275$$

Diferenças finitas com 5 pontos

$$f'(x) \simeq \frac{f(x-2h) - 8f(x-h) + 8f(x+h) - f(x+2h)}{12h}$$

$f(x) = 2x^2 + \text{sen}(\pi . x)$ $\quad f'(x) = 4x + \pi . \cos(\pi x)$

$h = 0.5; 0.25; 0.125$

$$f'(1) \simeq \frac{f(0) - 8f(0.5) + 8f(1.5) - f(2)}{12 \times 0.5} \simeq \frac{4}{3}$$

$$f'(1) \simeq \frac{f(0.5) - 8f(0.75) + 8f(1.25) - f(1.5)}{12 \times 0.25} \simeq 0.89543$$

$$f'(1) \simeq \frac{f(0.75) - 8f(0.875) + 8f(1.125) - f(1.25)}{12 \times 0.125} \simeq 0.86085$$

Valor exato $f'(1) = 0.8584$

$h = 0.5$	$f'(1) \simeq \frac{4}{3}$	$\frac{\varepsilon_a}{h^4} = 7.599$
$h = 0.25$	$f'(1) \simeq 0.89543$	$\frac{\varepsilon_a}{h^4} = 9.47968$
$h = 0.125$	$f'(1) \simeq 0.86085$	$\frac{\varepsilon_a}{h^4} = 10.0352$

Começa a estabilizar, logo confirma a ordem de precisão da fórmula $O(h^4)$

Derivada de segunda ordem num ponto

$$f''(x_0) \simeq \frac{1}{h^2} \times [f(x_0+h) - 2f(x_0) + f(x_0-h)]$$

Regra do Trapézio

$$\hat{I}_T(f) = \frac{b-a}{2}[f(a) + f(b)]$$

$$\int_1^2 x . e^x \, dx$$

$$\hat{I}_T(f) = \frac{2-1}{2}[f(1) + f(2)] = \frac{1}{2}(1 \times e^1 + 2 \times e^2) = \frac{1}{2} \times 17.496394 = 8.748197$$

Regra de Simpson

$$\hat{I}_S(f) = \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$

$$\int_1^2 x . e^x \, dx$$

$$\hat{I}_S(f) = \frac{2-1}{6}\left[f(1) + 4 \times f\left(\frac{1+2}{2}\right) + f(2)\right] = \frac{1}{6}\left[e^1 + 4 \times \frac{3}{2} . e^{3/2} + 2 . e^2\right] = 7.397755$$

Newton-Cotes Fechada de 2 pontos

↳ Regra do Trapézio

$$\int_0^1 e^{-\frac{x^2}{2}}\,dx$$

$$\hat{I}_T(f) = \frac{1-0}{2}[f(1)+f(0)] = \frac{1}{2}\left(e^{-\frac{1}{2}}+e^{-\frac{0}{2}}\right) = \frac{1}{2}\left(e^{-\frac{1}{2}}+1\right) \simeq 0.803265$$

Newton-Cotes Fechada de 3 pontos

↳ Regra de Simpson

$$\int_0^1 e^{-\frac{x^2}{2}}\,dx$$

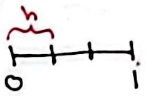
$$\hat{I}_S(f) = \frac{1-0}{6}\left[f(0)+4f\left(\frac{1+0}{2}\right)+f(1)\right] = \frac{1}{6}\left(1+4e^{-\frac{0.5^2}{2}}+e^{-\frac{1}{2}}\right) \simeq 0.856086$$

Newton-Cotes Fechada de 4 pontos

↳ Regra de Simpson dos 3 oitavos

$$\hat{I}(f) = \frac{3h}{8}\times\left[f(x_0)+3f(x_1)+3f(x_2)+f(x_3)\right]$$

$$\int_0^1 e^{-\frac{x^2}{2}}\,dx$$



$h = \frac{1}{3}$

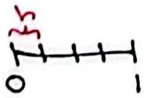
$$\hat{I}(f) = \frac{3\times 1/3}{8}\times\left[f(0)+3f(1/3)+3f(2/3)+f(1)\right] =$$

$$= \frac{1}{8}\left[1+e^{-\frac{(1/3)^2}{2}}\times 3+3\times e^{-\frac{(2/3)^2}{2}}+e^{-\frac{1}{2}}\right] \simeq 0.855828$$

Newton-Cotes Fechada de 5 pontos

$$\hat{I}(f) = \frac{2h}{45}\times\left[7f(x_0)+32f(x_1)+12f(x_2)+32f(x_3)+7f(x_4)\right]$$

$$\int_0^1 e^{-\frac{x^2}{2}}\,dx$$



$h = \frac{1}{4}$

$$\hat{I}(f) = \frac{2\times 1/4}{45}\times\left[7+32\times e^{-\frac{(1/4)^2}{2}}+12e^{-\frac{(1/2)^2}{2}}+32e^{-\frac{(3/4)^2}{2}}+7e^{-\frac{1}{2}}\right] \simeq 0.855622$$

Trapézio Composta

$$\hat{I}_T(f) = \frac{h}{2}[f(a)+f(b)]+h\times\sum_{k=1}^{n-1} f(x_k)$$

Simpson Composta

$$\hat{I}_S(f) = \frac{h}{3}\left[f(a)+f(b)+4\sum_{k=1}^{m} f(x_{2k-1})+2\sum_{k=1}^{m-1} f(x_{2k})\right]$$

Regra do Ponto Médio

$$\hat{I}(f) = (b-a)f\left(\frac{a+b}{2}\right)$$

Newton-Cotes Aberta de 2 pontos

$$\hat{I}(f) = \frac{3h}{2}\left[f(x_0)+f(x_1)\right]$$

Newton-Cotes Aberta de 3 pontos

$$\hat{I}(f) = \frac{4h}{3}\left[2f(x_0)-f(x_1)+2f(x_2)\right]$$

Newton-Cotes Aberta de 4 pontos

$$\hat{I}(f) = \frac{5h}{24}\left[11f(x_0)+f(x_1)+f(x_2)+11f(x_3)\right]$$

Integração de Romberg

↳ Simpson Composta: $\hat{I}_S^{(r)}(h) = \dfrac{4^{r+1}\cdot \hat{I}_S^{(r-1)}(h/2) - \hat{I}_S^{(r-1)}(h)}{4^{r+1}-1}$ ⊛

↳ Trapézio Composta: $\hat{I}_T^{(r)}(h) = \dfrac{4^{r}\cdot \hat{I}_T^{(r-1)}(h/2) - \hat{I}_T^{(r-1)}(h)}{4^{r}-1}$

$$\int_0^1 \frac{1}{1+x^2}\,dx \qquad \hat{I}_T^{(3)}(h) = ?$$

$O(h^2)$	$O(h^4)$	$O(h^6)$	$O(h^8)$
$\hat{I}_T(h)$			
$\hat{I}_T(h/2)$	$\hat{I}_T^{(1)}(h)$		
$\hat{I}_T(h/2^2)$	$\hat{I}_T^{(1)}(h/2)$	$\hat{I}_T^{(2)}(h)$	
$\hat{I}_T(h/2^3)$	$\hat{I}_T^{(1)}(h/2^2)$	$\hat{I}_T^{(2)}(h/2)$	$\hat{I}_T^{(3)}(h)$

$h = b-a = 1$

Trapézio Simples

$$\hat{I}_T(h) = \frac{h}{2}\left[f(a)+f(b)\right] = \frac{1}{2}\left[f(0)+f(1)\right] = 0.750000$$

Trapézio Composta

$$\hat{I}_T(h/2) = \frac{1/2}{2}\left[f(0)+f(1)\right] + \frac{1}{2}\left[f(0.5)\right] = 0.775000$$

$$\hat{I}_T(h/2^2) = \frac{1/2^2}{2}\left[f(0)+f(1)\right] + \frac{1}{4}\left[f(0.25)+f(0.5)+f(0.75)\right] = 0.782794$$

$$\hat{I}_T(h/2^3) = \frac{1/2^3}{2}\left[f(0)+f(1)\right] + \frac{1}{8}\left[f(0.125)+f(0.25)+f(0.375)+f(0.5)+f(0.625)+ \right.$$
$$\left. + f(0.75)+f(0.875)\right] = 0.784747$$

⊛

(*)

r=0	r=1	r=2	r=3
0.75			
	0.783333		
0.775		0.785529	
	0.785392		0.785396
0.782794		0.785399	
	0.785398		
0.784747			

$$\hat{I}_T^{(1)}(h) = \frac{4^1 . \hat{I}_T(h/2) - \hat{I}_T(h)}{4^1 - 1} = \frac{4 \times 0.775 - 0.75}{3} = 0.783333$$

$$\hat{I}_T^{(1)}(h/2) = \frac{4^1 . \hat{I}_T(h/2^2) - \hat{I}_T(h/2)}{4^1 - 1} = 0.785392$$

$$\hat{I}_T^{(1)}(h/2^2) = \frac{4^1 . \hat{I}_T(h/2^3) - \hat{I}_T(h/2^2)}{4^1 - 1} = 0.785398$$

$$\hat{I}_T^{(2)}(h) = \frac{4^2 . \hat{I}_T^{(1)}(h/2) - \hat{I}_T^{(1)}(h)}{4^2 - 1} = \frac{16 \times 0.785392 - 0.783333}{15} = 0.785529$$

$$\hat{I}_T^{(2)}(h/2) = \frac{4^2 . \hat{I}_T^{(1)}(h/2^2) - \hat{I}_T^{(1)}(h/2)}{4^2 - 1} = 0.785399$$

$$\hat{I}_T^{(3)}(h) = \frac{4^3 . \hat{I}_T^{(2)}(h/2) - \hat{I}_T^{(2)}(h)}{4^3 - 1} = \frac{64 \times 0.785399 - 0.785529}{63} = 0.785396$$